

```

getLogger(MySpringApp.class);

public static void main(String[] args) {
    SpringApplication.run(MySpringApp.class, args);
    logger.info("Spring application started!");
}
}

```

The assistant does not wash its hands off with just code. It gives a lot of notes as well. For instance, the following is the note generated along with the above code.

In this example, we use the `LoggerFactory` class from `slf4j` to create a logger for the `MySpringApp` class. We can then use the logger object to output log statements at different levels (e.g., info, debug, error). To use `slf4j` in a Spring application, you will need to include the `slf4j-api` library as a dependency in your project. You will also need to include a concrete implementation of `slf4j`, such as `log4j`, `logback`, or `java.util.logging`. For example, to use `log4j` as the underlying logging implementation, you will need to include the `log4j` library as a dependency in your project. You can then configure `log4j` to output log statements to the desired destination (e.g., console, file).

Since it is advising some dependencies, I acted as if I do not know how it can be done. So, I asked this follow-up question:

pom file for spring with slf4j

Here is the resulting suggestion:

```

<project>
  <!-- Other project configuration -->

  <dependencies>
    <!-- Other dependencies -->
    <dependency>
      <groupId>org.slf4j</groupId>
      <artifactId>slf4j-api</artifactId>
      <version>1.7.30</version>
    </dependency>
    <dependency>
      <groupId>org.slf4j</groupId>
      <artifactId>slf4j-log4j12</artifactId>
      <version>1.7.30</version>
    </dependency>
  </dependencies>
</project>

```

Again, the notes suggested a `log4j` dependency. I asked the assistant to add that too, and it generated the following refined configuration.

```

<project>
  <!-- Other project configuration -->

  <dependencies>
    <!-- Other dependencies -->
    <dependency>
      <groupId>org.slf4j</groupId>
      <artifactId>slf4j-api</artifactId>
      <version>1.7.30</version>
    </dependency>
    <dependency>
      <groupId>org.slf4j</groupId>
      <artifactId>slf4j-log4j12</artifactId>
      <version>1.7.30</version>
    </dependency>
    <dependency>
      <groupId>log4j</groupId>
      <artifactId>log4j</artifactId>
      <version>1.2.17</version>
    </dependency>
  </dependencies>
</project>

```

You will also need to include a `log4j.properties` file in your project to configure `log4j` to output log statements to the desired destination (e.g., console, file).

The story doesn't end here. I asked the assistant to help me with the `log4j.properties` file:

log4j.properties for spring with slf4j

It obliged.

```

# Root logger
log4j.rootLogger=INFO, stdout, file

# Console appender
log4j.appender.stdout=org.apache.log4j.ConsoleAppender
log4j.appender.stdout.layout=org.apache.log4j.PatternLayout
log4j.appender.stdout.layout.ConversionPattern=%d{yyyy-MM-dd HH:mm:ss} %-5p %c{1}:%L - %m%n

# File appender
log4j.appender.file=org.apache.log4j.FileAppender
log4j.appender.file.File=app.log
log4j.appender.file.layout=org.apache.log4j.PatternLayout
log4j.appender.file.layout.ConversionPattern=%d{yyyy-MM-dd HH:mm:ss} %-5p %c{1}:%L - %m%n

```